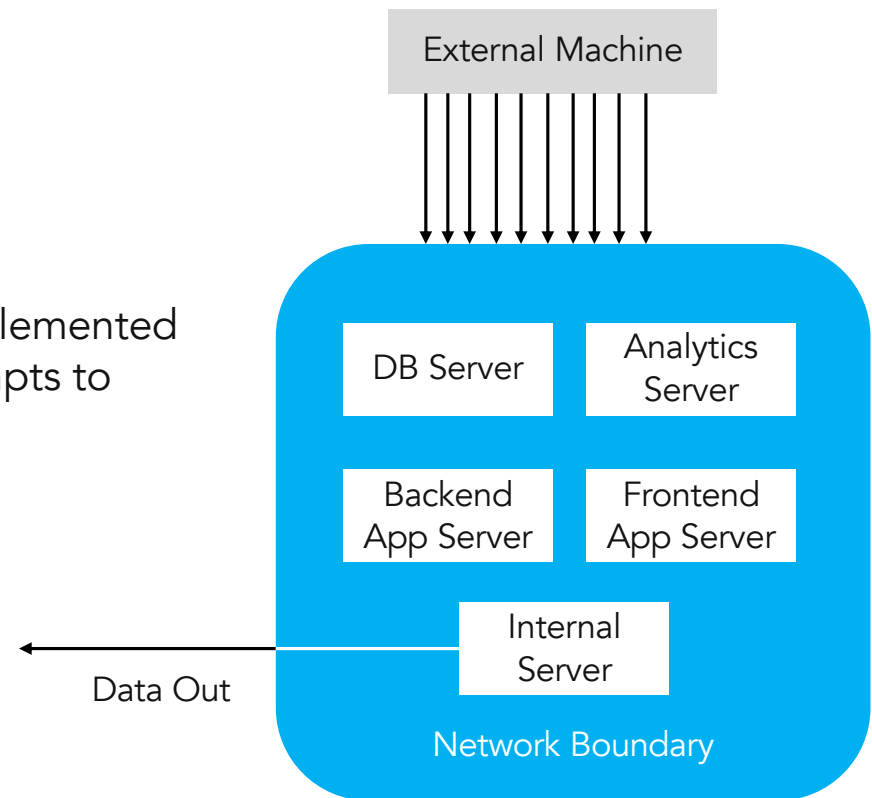


# Intrusion Detection

[www.3aayaam.in](http://www.3aayaam.in)

# Intrusion Detection

- Attempt to get into a system to extract sensitive information by a malicious agent.
- Network Intrusion.
- Host Intrusion.
- Intrusion Detection is a security feature that is implemented in every organization to detect and mitigate attempts to compromise the digital system.
- Network Intrusion Detection (NIDS)
- Brute force attack
- Data exfiltration
- Host Intrusion Detection (HIDS).



# Intrusion Detection

- Network Intrusion Detection (NIDS) - Unauthorized activity on a digital network. It is a digital break-in where someone bypasses security to gain access to data, disrupt operations, or plant malicious software.
- Brute force attack - Trial-and-error method to guess login credentials, encryption keys, or hidden. Rather than using sophisticated exploits, this attack relies on computing power to churn through thousands (or millions) of variations until the correct one is found.
- Data exfiltration - Unauthorized transfer or delete of sensitive information from a system. In this case, data actually leaves the digital premises.
- Host Intrusion Detection (HIDS) - host intrusion is the successful, unauthorized entry into a specific computer or server.

# Network Intrusion Detection

- NIDS should be handled in real-time.

NwDataTimestamp  
FlowID  
SrcIP  
DstIP  
SrcPort  
DstPort  
Protocol  
DataPackets  
DataBytesIn  
DataBytesOut  
Action  
UserID

Ingest

nw\_data\_flow\_data\_day\*.csv

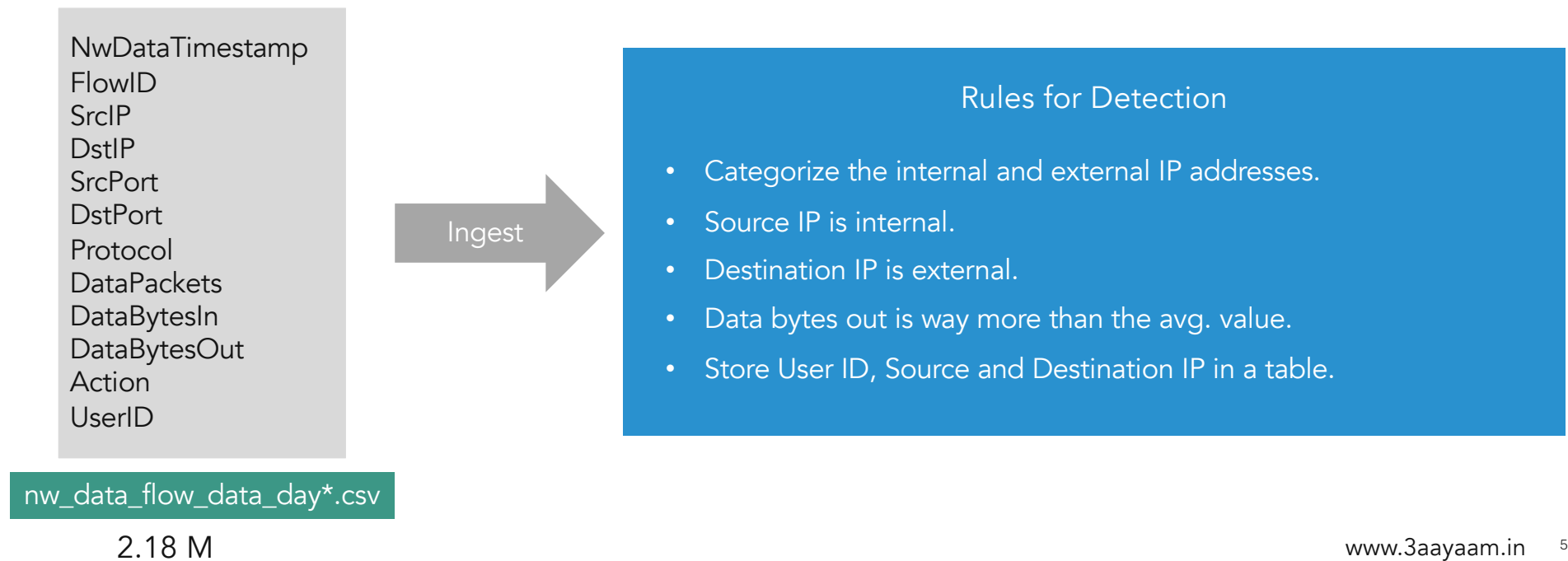
2.18 M

## Rules for Detection

- Single source IP is targeting single destination IP continuously. Every time the access is being denied.
- Real time output should monitor such activity over a period of 1 min.
- The IP address, Protocol, Time Range, Duration of attack, No. of attempts, No. of data packets, User ID should be added to a BLOCKLIST table.
- Which protocol & port combination is getting used the most and which server is being targeted the most, should be stored.
- Is the attacker trying send some information/data during attempt?

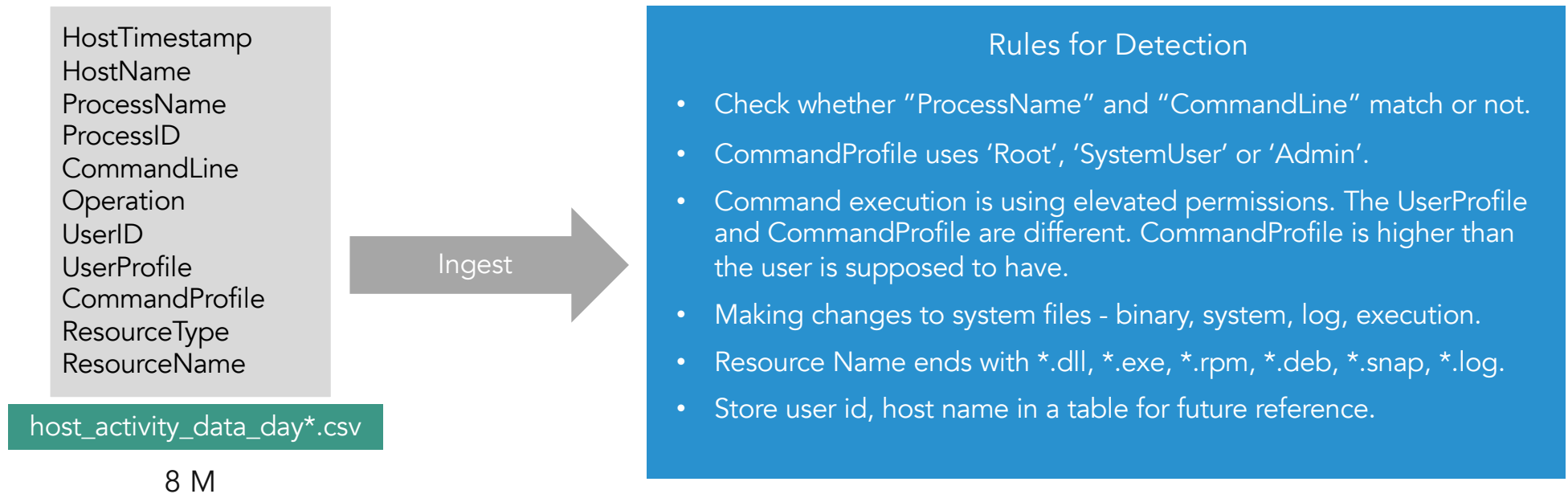
# Data Exfiltration

- Data exfiltration analysis should process data every 30 mins.



# Host Intrusion Detection

- Host intrusion detection analysis should run once in every 1-2 hours.



# Brute Force Attack

- Brute force attack analysis should be run once a day.

